

For the week of September 22-26

1. Integration by substitution tells us that $\int f'(u(x))u'(x)dx = f(u(x)) + C$. For each of the following integrals, identify $f(x)$, $u(x)$ and $u'(x)$, then evaluate the integral. You may need to rewrite the integral in a different form to identify $u'(x)$.

(a) $\int t(t^2 - 8)^{10} dt$

(b) $\int xe^{2x^2} dx$

(c) $\int \frac{y}{y^2+4} dy$

2. Evaluate the following integrals using substitution.

(a) $\int (x+5)^2 dx$

(b) $\int \frac{e^x}{2+e^x} dx$

(c) $\int \sqrt{1+\sqrt{x}} dx$

(d) $\int (x+8)(3-4x)^{1/3} dx$

3. Evaluate the following definite integrals using substitution.

(a) $\int_0^{3/2} \cos(\pi\theta) d\theta$

(b) $\int_1^4 \frac{1}{t+4} dt$

(c) $\int_{-1}^1 \frac{e^x}{2+e^x} dx$

4. In the following, find a function $g(x)$ so that the integral may be evaluated using substitution. You don't need to evaluate the integral.

(a) $\int g(x)e^{x^4} dx$

(b) $\int g(\theta)\cos^2(\theta) d\theta$

5. In this problem we will calculate $\int \sin(\theta)\cos(\theta) d\theta$ several different ways.

- (a) First, calculate this integral by making a substitution.
 - (b) Next, calculate it using the *other* substitution (if you used $u = \sin(\theta)$ before, now use $u = \cos(\theta)$).
 - (c) Use the trigonometric identity $\sin(2\theta) = 2\sin(\theta)\cos(\theta)$ to compute this integral another way.
 - (d) Compare your answers with one another. Do they agree?
6. Show the following integrals are equal using a substitution.
- (a) $\int_0^{\pi/3} 3\sin^2(3x)dx = \int_0^{\pi} \sin^2(y)dy$
 - (b) $\int_1^e (\ln x)^3 dx = \int_0^1 y^3 e^y dy$
7. Evaluate the following integrals using integration by parts.
- (a) $\int \ln(x)dx$
 - (b) $\int x^3 \ln(x)dx$
 - (c) $\int \sin^2(\theta)d\theta$
 - (d) $\int \frac{z}{e^z} dz$
 - (e) $\int \theta^2 \cos(\theta)d\theta$
8. Evaluate the following integrals using integration by parts. You may need to use integration by parts several times.
- (a) $\int xe^x dx$
 - (b) $\int x^2 e^x dx$
 - (c) $\int x^3 e^x dx$
 - (d) (Harder; will not be on a quiz) Write a formula for $\int x^n e^x dx$ where $n = 0, 1, 2, \dots$. Your formula may use summation notation and should not have any integrals in it.

9. Evaluate the following definite integrals using integration by parts.

(a) $\int_1^5 \ln(t) dt$

(b) $\int_1^3 x \ln(x) dx$

10. In the following integrals, find a function $g(x)$ so that the integral may be evaluated using integration by parts *once*.

(a) $\int g(x)e^x dx$

(b) $\int x^2 g(x) dx$

(c) $\int x^3 \sin(g(x)) dx$

11. Use integration by parts to write the following expression in a form which has no integrals. Your answer will be in terms of f .

$$\int f''(x) \ln(x) dx + \int \frac{f(x)}{x^2} dx.$$

12. Evaluate the following integrals.

(a) $\int \frac{1}{\sqrt{4x-1}} dx$

(b) $\int x^3 \cos(x^2) dx$

(c) $\int_0^1 \arctan(z) dz$ (recall $\frac{d}{dz} \arctan(z) = \frac{1}{1+z^2}$).

(d) $\int \frac{\ln((\ln(x))^3)}{x} dx$

The following problems are for practice only and will not appear on quizzes.

I) Textbook section 7.1, problems # 3-50, 59-66.

II) Textbook section 7.2, problems # 3-37, 70-73, 88-90.